Francisco Rowe

Geographic Data Science Lab
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Research highlights

Research output: 100+ scientific journal articles; 5,168 Google Scholar citations and h-index 36 (23 October 2026).

Citation recognition: Listed among the world's top-cited scientists in the Stanford/Elsevier database for single-year citation impact in 2025 (2026 release).

Research funding: Principal or co-investigator on projects with a combined funding value of £12.8 million.

International distinction: Recipient of the 2026 International Geographical Union Award for Distinguished Geographical Practice.

Policy leadership: Co-Vice Chair of the UN Mobile Phone Data Task Team and lead of its Migration Statistics Subgroup.

Doctoral supervision: 17 PhD students supervised to completion.

International policy engagement: Part of Wilton Park's international policy network, bringing together leading experts and decision-makers to address global challenges. Wilton Park is an executive agency of the UK Foreign, Commonwealth & Development Office.

RESEARCH INTERESTS

My research focuses on human mobility and migration, spatial inequality, geographic data science and climate sustainability. I use digital trace data and computational methods to study the causes and impacts of population movements, particularly during climate hazards, health emergencies and conflict. My work also examines how urban form shapes mobility and exposure to climate risks. I develop methods that make rapidly collected data more useful for population research, official statistics and policy. I lead the Geographic Data Science Lab and am Co-Deputy Director of the MRC Social Health Hub.

EDUCATION

Ph.D. 2013 - Economic Geography, University of Queensland, Brisbane, Australia.

M.S. 2007 - MSc in Regional Science, Universidad del Norte, Antofagasta, Chile.

B.S. 2006 - BA in Business Management with specialisation in Economics, *summa cum laude*, Universidad del Norte, Antofagasta, Chile.

PROFESSIONAL TRAINING AND QUALIFICATIONS

2023 Heilbron Leadership Programme. The University of Liverpool, Liverpool, United Kingdom.

2018 Certificate in Professional Studies in Learning and Teaching in Higher Education (CPS) - Distinction. The University of Liverpool, Liverpool, United Kingdom.

2014 Spatial Scientist. VETASSESS, Brisbane, Australia.

APPOINTMENTS

Current appointments

2023– Professor in Population Data Science
Department of Geography and Planning, University of Liverpool, Liverpool, United Kingdom

2020– Lead of the Geographic Data Science Lab
Department of Geography and Planning, University of Liverpool, Liverpool, United Kingdom

Honorary appointments

2020– Visiting Professor
National Tsing Hua University, Taiwan.

2021– Visiting Professor
Data Science Institute, Universidad del Desarrollo, Santiago, Chile.

Distinguished Leadership and Advisory Appointments

2025– ONS Advisor
UK Office of National Statistics (ONS), Working Group on Internal Migration Methods Development, London, United Kingdom.

2025– ESRC Peer Review College member
Economic Social Research Council (ESRC) Peer Review College, London, United Kingdom.

2024– Co-Vice Chair
Mobile Phone Data Task Team of the UN Committee of Experts on Big Data and Data Science for Official Statistics, UN Big Data, Statistics Division, New York, United States.

2024– Lead
Migration Statistics Subgroup, Mobile Phone Data Task Team of the UN Committee of Experts on Big Data and Data Science for Official Statistics, UN Big Data, Statistics Division, New York, United States.

2024– Co-Deputy Director
Social Health Hub, Medical Research Council, United Kingdom.

2022– Chair
Quantitative Research Methods Group, The Royal Geographical Society with the Institute of British Geographers, United Kingdom.

2022– Secretary
Steering Committee of the International Geographical Union (IGU) Commission on Population Geography, United Kingdom.

Past appointments

2018–23 Senior Lecturer in Quantitative Human Geography
Department of Geography and Planning, University of Liverpool, Liverpool, United Kingdom

2018–22 Office of National Statistics, Government Statistical Service Advisory Committee. Use of administrative data.

2018–22 Institutional Lead of the Longitudinal Data Analysis Specialised Training Route
Economic and Social Research Council - North West Social Science Doctoral Training Partnership, United Kingdom.

2021–22 Events Officer
Population Geography Research Group, The Royal Geographical Society with the Institute of British Geographers, United Kingdom.

2016–18 Lecturer in Quantitative Human Geography
Department of Geography and Planning, University of Liverpool, Liverpool, United Kingdom

2013–16 Postdoctoral Research Fellow
School of Geography, Planning and Environmental Management, The University of Queensland, Brisbane, Australia.

2014–16 ResTeach Fellow
School of Geography, Planning and Environmental Management, The University of Queensland, Brisbane, Australia.

2010–13 University Tutor
School of Geography, Planning and Environmental Management; The University of Queensland, Brisbane, Australia.

2010–13 Research Assistant
Queensland Centre for Population Research, School of Geography, Planning and Environmental Management, The University of Queensland, Brisbane, Australia.

2010–11 Research Assistant
Queensland Centre for Population Research, School of Geography, Planning and Environmental Management, The University of Queensland, Brisbane, Australia.

2008–09 Research Officer
Millennium Nucleus for Regional Science and Public Policy. The Chilean Ministry of Planning and Universidad Católica del Norte, Antofagasta, Chile.

2007–08 Research Analyst
CONYMET, Antofagasta, Chile.

2007–08 Lecturer (Part-Time),
Faculty of Business and Economics, Department of Economics, Universidad Católica del Norte, Antofagasta, Chile.

2007 Lecturer (Part-Time),
School of Business Management, Universidad Santo Tómas, Antofagasta, Chile.

EDITORIAL ROLES

2024– Editorial board member
Population, Space and Place.

2023– Editorial board member
Spatial Economic Analysis.

2025– Editorial board member
Applied Spatial Analysis and Policy.

2022–26 Managing Editor
REGION. The Journal of the European Regional Science Association.

2025–26 Guest Editor
Special Issue: Migration and Sustainable Urbanisation in the Global South.
Co-editors: Hengyu Gu, Chenglong Wang, Jianfa Shen, Francisco Rowe and Ye Liu.

2023–25 Guest Editor
Special Issue: CATMOG2.0: ComputAtional noTebooks to enable MOdern Geographic data analysis. REGION. The Journal of the European Regional Science Association.
Co-editors: Gunther Maier, Levi Wolf, Qunshan Zhao, Caitlin Robinson, Alessia Calafiore, Elisabetta Pietostefani and Stephen Law.

2023–24 Guest Editor
Special Issue: GeoAI for Urban Sensing. Urban Informatics.
Co-editors: Yuhao Kang, Arianna Salazar-Miranda, Yiyi He and Song Gao.

2023–24 Guest Editor
Special Issue: Sustainable spatiotemporal analytics with repeatable, reproducible, and expandable (RRE) framework and design. International Journal of Applied Earth Observation and Geoinformation.
Co-editors: Siqin Wang, Xiao Huang, Filip Bijjecki and Veruska Muccione.

2021–23 Social Media Editor
Journal of Royal Statistical Society – Series A

2022–23 Guest Editor
Special Issue: Internal Migration in Times of COVID-19, Population, Space and Place.
Co-editors: Miguel Gonzalez-Leonardo and Tony Champion.

2018–21 Associate Editor
REGION. The Journal of the European Regional Science Association.

2019–20 Guest Editor
Special Issue: The Potential of Notebooks for Scientific Publication: Reproducibility, and Dissemination. REGION. The Journal of the European Regional Science Association.
Co-editors: Dani Arribas-Bel, Gunther Maier and Serge Rey.

2015–17 Guest Editor
Special Issue: Youth and graduate migration, The Annals of Regional Science.
Co-editors: Alessandra Faggian and Jonathan Corcoran.

COMMITTEE MEMBERSHIPS

2025 Scientific Committee. The Sixth Spatial Data Science Symposium. December 4-5, 2025, Distributed and online.

2025 Scientific Committee. Regional Science in Turbulent Times. In search of a resilient, sustainable and inclusive future. European Regional Science Association Conference 2025, Athens, Greece.

2024 Programme Committee. Biases and Representativeness in Digital Forms of Spatiotemporal Data Panel Session. Symposium on Spatiotemporal Data Science 2024, Virginia Tech Research Center – Arlington & George Mason Univ. Mason Square July 22-24, 2024.

2024 Scientific Committee. Regional Science Dialogues for Peace and Sustainable Development. European Regional Science Association Conference 2024, Azores, Portugal.

2023 Scientific Committee. Urban Challenges and Sustainable Technological Revolution. European Regional Science Association Conference 2023, Alicante, Spain.

2022 Programme Committee. Migration and Mobility Research in the Digital Era. Conference on Complex Systems 2022, Palma de Mallorca, Spain.

2022 Scientific Committee. Disparities in a Digitalising (Post-Covid) world – Networks, Entrepreneurship and Regional Development. European Regional Science Association Conference 2023, Pécs, Hungary.

2021 Programme Committee. Spatial and Temporal Thinking in Data-Driven Methods. Spatial Data Science Symposium 2021.

2021 Programme Committee. The 2nd international workshop on Migration and Mobility Research in the Digital Era. Satellite event of the Conference on Complex Systems 2021, Lyon, France.

2020–22 Spatial Inequality Expert Advisor. Teesside Task Force. The UK2070 Commission.

2020–22 Use of Administrative Data Expert Advisor. Office of National Statistics (ONS) Government Statistical Service Advisory Committee.

2018–19 Migration Expert Advisor. RECH Contribution Committee. Repayment mechanisms of Chile government's international postgraduate scholarship programme.

2018 The North American Regional Science Council (NARSC) Graduate Student Paper Competition Committee. San Antonio, Texas.

FUNDING AND AWARDS

Grants

Principal or co-investigator on projects with a combined funding value of £12.8 million.

2025 "Joint Data Products Linking Satellite, Synthetic and Trajectory Data". Economic and Social Research Council (ESRC). PI £149,779.54.

2024 "IMAGO - The SDR UK Imagery Data Service for Sustainability, Prosperity and Wellbeing". Economic and Social Research Council (ESRC). PI £6,788,643.

2024 "Dynamic Estimates of Displacement in Conflict Regions: A Policy-driven framework triangulating data". Research England. Co-I £49,509.99.

2024 "Harnessing Satellite and GPS Phone Data for Policy-Driven Assessment of Internal Population Displacement in Ukraine". Research England. Co-I £9,244.70.

2024 "DEBIAS - Developing a framework to measure and correct biases in human mobility data extracted from digital footprints". Economic and Social Research Council (ESRC). PI £188,150.00.

2024 "Social Health Hub - Building recovery and resilience in severe mental illness: Leveraging the

role of social determinants in illness trajectories and interventions". Medical Research Council. PI £4,346,865.

2023 "TIMELY - Validating an up-to-date dataset of internal migration to inform local and regional-scale spatial planning strategies in Chile". Public Policy Quality-Related Funding. Research England. PI. £74,136.53

2022 "RECAST - Strengthening policy collaboration with the United Nations: Recovery and resilience through digital footprint data". Public Policy Quality-Related Funding. Research England. PI. £89,913.60

2022 "Healthcare Communications and Information Distortion in Crisis Situations: The Case of COVID-19 in Merseyside". Aviva / The Pandemic Institute: Recover Funding Call. Co-I. £49,999.10.

2022 "Digital Media & Society Institute". University of Liverpool. Co-I. £169,881.

2022 "Supporting Post Pandemic Recovery and Resilience through New Forms of Data". Economic and Social Research Council (ESRC). AHRC UK South Korea SSH Connections Grants. Co-I. £138,893.36.

2021 "InequaliTies IN Experiencing uRbAn fuNcTion (ITINERANT) During COVID-19". Engineering and Physical Sciences Research Council (EPSRC) Impact Acceleration Accounts PI. £16,032.

2021 "InequaliTies IN Experiencing uRbAn fuNcTion (ITINERANT)". Engineering and Physical Sciences Research Council (EPSRC) / The Alan Turing Institute. Co-I. £128,499.

2021 "Understanding Anti-Asian Sentiment During the COVID-19 Pandemic". Research England. Public Policy Quality-Related Strategic Priorities Fund. PI. £25,000.

2021 "Assessing the Geography and Impacts of Socio-economic Inequalities in the UK". Research England. Public Policy Quality-Related Strategic Priorities Fund. PI. £15,000.

2020 "Generating actionable evidence for containing the spread of misinformation". Liverpool COVID-19 Partnership Strategic Research Fund. Co-I. £20,100.

2020 "Misinformation and COVID-19 in social media in Venezuela." Global Challenges Strategy Group. Co-I. £5,000.

2020 "Supporting evidence-based policy actions". Research England. Public Policy Quality-Related Strategic Priorities Fund. University of Liverpool. PI. £20,000.

2019 "Understanding and Predicting the Long-term Labour Market and Migration Trajectories of Immigrants and Their Children in the United Kingdom" (ES/P000665/1). Economic and Social Research Council (ESRC). PI. £135,400.

2017 "Past and future of urban structures in Latin American cities". ODA Seed Fund Project, University of Liverpool. Co-I. £8,900.

2016 "Understanding the declining trend in internal migration in Europe". Regional Studies Association (RSA). PI. £33,490.39.

2016 "4th Chilean Graduate Conference: Advanced knowledge to enhance Chile's development". Council on Australia Latin America Relations (COALAR) project. PI. AU$10,000 (£5,600).

2016 "Application of advanced knowledge to enhance Chile's development". ChileGlobal Project. PI. AU$22,500 (£12,569.86).

2014 "Improving Subnational Population Forecasts". Australian Research Council (ARC) Discovery Project (DP150103343). PI. AU$ 293,980 (£158,150.39).

2014 "ResTeach grant". School of Geography Planning and Environmental Management. University of Queensland. PI. AU$10,000 (£5,600).

2014 "UQ support grant for Engagement and Internationalisation". School of Geography Planning and Environmental Management. University of Queensland. PI. AU$1,620 (£905).

2013 "ResTeach grant". School of Geography Planning and Environmental Management. University of Queensland. PI. AU$10,000 (£5,600).

2012 "Doctoral studentship: Modelling the spatial patterns of labour mobility in Chile". National Commission for Scientific and Technological Research (CONICYT). Government of Chile. PI. £140,000.

2007 "Business Platform for promoting innovation in the mining sector in Chile", CONYMET Ltda and The Chilean Economic Development Agency (CORFO)-INNOVA. Project ID: 07DPN-6752. PI. CLP$30,000,000 (£28,236.57)

Consultancy

2024 "A short audit review of work which has been completed for the Department for Levelling Up, Housing and Communities on tall buildings in the UK". Livewire Consultancy Ltd . https://www.livewireconsultancy.com

2024 "Geographic data science and modelling - skill training capacity" Liverpool City Region Combined Authority. https://www.liverpoolcityregion-ca.gov.uk

2021 "What do "left behind" areas look like over time?" Let Behind Neighbourhoods Local Trust. https://localtrust.org.uk

2020 "Analysis of Twitter data to measure anti-migration sentiment during COVID- 19". International Organization for Migration. Global Migration Data Analysis Centre (GMDAC). https://gmdac.iom.int

2020 "Data science training programme". The Universities and Colleges Admissions Service (UCAS). https://www.ucas.com

Research recognition

2026 Listed among the world's top-cited scientists in the Stanford/Elsevier database, single-year citation impact for 2025. August 2026 update, version 9, published 7 October 2026. doi.org/10.17632/btchxktzyw.9

Awards

2026 "International Geographical Union Award for Distinguished Geographical Practice". International Geographical Union.

2025 "Staff Awards - Nomination to Inspiration Award". School of Environmental Sciences.

2024 "Staff Awards - Nomination to Inspiration Award". School of Environmental Sciences.

2022 "2022 Philip Leverhulme Prize Nomination in Geography". Leverhulme Trust.

2022 "Staff Awards - Outstanding contribution to Research". School of Environmental Sciences.

2022 "Staff Awards - Partnership of the year, University of Liverpool Awards". Finalist shortlisting - Geographic Data Science Lab.

2022 "Staff Awards - Research Impact of the Year, University of Liverpool Awards". Finalist shortlisting - Geographic Data Science Lab.

2021 "Journal of Geographical Systems Editors' Choice Award". Best article: Projecting the demographic impact of Syrian migration in a rapidly ageing society, Germany. Special Issue "Spatial Demography in Regional Science". doi.org/10.1007/s10109-021-00357-3

2018 "Best Paper Spatial Economic Analysis". An evolutionary approach to the delimitation of labour market areas: an empirical application for Chile. Regional Studies Association. doi.org/10.1080/17421772.2017.1273541

2014 "Dean's Award for Research Higher Degree Excellence". University of Queensland. Fewer than 10% of PhD graduates from the University of Queensland are recognised this way each year.

2007 "Best educational performance award, 2002 Business Management cohort". Universidad Catolica del Norte.

PUBLICATIONS

Journal Articles

I have published over 100 scientific articles. As of 23 October 2026, my Google Scholar profile records 5,168 citations and an h-index of 36.

2026 Neville, R., Rowe, F., Zagheni, E. "Machine learning forecasts suggest a concentration paradox in international student mobility to the United Kingdom". Nature Communications 17, 9604. doi.org/10.1038/s41467-026-77425-z

2026 Cabrera, C., Rowe, F. "Making hidden biases visible in population location data from mobile phones". Royal Society Open Science 13(8), 251703. doi.org/10.1098/rsos.251703

2026 Mahony, M., Rowe, F. "A Biographical Approach to Understand Long-Term Occupational Trajectories of Immigrants in the UK Labour Market". Population, Space and Place 32(6), e70308. doi.org/10.1002/psp.70308

2026 Nasuto, A., Iacus, S., Rowe, F. Jain, D., "Large language models identify immigration attitudes in online discourse regardless of language". Scientific Reports doi.org/10.1038/s41598-026-52167-6

2026 Mahony, M., Rowe, F. "Urban enclaves have contrasting labour-market effects for migrants and ethnic minorities in England". Cities doi.org/10.1016/j.cities.2026.106833

2026 Fierro, P., Rodríguez-Pose, A., Rowe, F., Helsper, E. "Voting Red Again: How Social Capital And Local Change Drove The Trump Swing". Journal of Regional Science. doi.org/10.31219/osf.io/fuajw

2026 Rowe, F. "Age-verification is skewing human location data-we must safeguard this vital resource". OFSdoi.org/10.31235/osf.io/fz7h9_v1

2026 Wang, S., Huang, X., Biljecki, F., Rowe, F., Muccione, V. "Advancing sustainable geospatial analytics and geoinformatics through repeatable, reproducible, and expandable (RRE) framework and design". International Journal of Applied Earth Observation and Geoinformation doi.org/10.1016/j.jag.2026.105239

2026 Berzi, M., Albertini M., Barranco R., Brunet-Jailly E., Chilla T., Dijkstra L., Duvernet C., Finnsdottir M., Galic A., Gareis P., Gaugitsch R., Günther E., Hallgrimsdottir H., Herrmann

B., Hippe S., Jacobs-Crisioni C., Jánosi V., Järv O., Kompil M., Kučas A., Paszto V., Poorthuis A., Rowe F., Rubio J., Van De Weghe N., Van Der Valk J., Ferreira R. "Mapping flows, shaping spaces: a review of data-driven approaches to EU cross-border regions". European Planning Studies doi.org/10.1080/09654313.2026.2628289

2026 Mason, M., Rowe, F. "Tweeting migration: A decade of shifting sentiment across local areas of Great Britain". Journal of Ethnic and Migration Studies. doi.org/10.1080/1369183X.2026.2700232

2025 Iradukunda, R., Rowe, F., Pietrostefani, E. "Estimating internal displacement in Ukraine from high-frequency GPS mobile phone data". Nature Humanities and Social Sciences Communications doi.org/10.1057/s41599-025-06137-4

2025 González-Leonardo, M., Newsham, N., Zazueta-Borboa, D., Baptista, E., Rowe, F. "Assessing the accuracy of ARIMA models to forecast population at the intraurban level in Greater Mexico City". Population Research and Policy Review doi.org/10.31235/osf.io/ngkzr

2025 González-Leonardo, M., Cabrera, C., Neville, R., Nasuto, A., Rowe, F. "Beyond the Immediate Impacts of COVID-19 on Internal Population Movements in Mexico: Facebook Data Reveal Urban Decay and Slow Recovery—A Research Note". Demography. doi.org/10.1215/00703370-12183205

2025 Rodríguez-Puello, G., Hernandez, L., Romero-Espinosa, D., Rowe, F. "The disruptive long-term costs of international migration on subjective well-being". Journal of Happiness Studies. doi.org/10.1007/s10902-025-00916-1

2025 Newsham, N., Rowe, F. "The Demographic Causes of European Sub-National Population Declines". European Journal of Population. doi.org/10.1007/s10680-025-09730-0

2025 Frazier, A.E., Nelson, T., Kedron, P., Shook, E., Dodge, S., Murray, A., Goodchild, M., Battersby, S., Blanford, J.I., Claramunt, C., Holler, J., Koylu, C., Lee, A., Manson, S., Şalap-Ayça, S., Wilson, J.P., Zhao, B., Bennett, L., Cabrera, C., Franklin, R., McKenzie, G., Miller, H., Oshan, T., Rey, S., Rowe, F., Spielman, S. and Xu, W. "Rethinking GIScience education in an age of disruptions". Transactions in GIS. doi.org/10.1111/tgis.70048

2025 Ward, C., Singleton, A., Robinson, C., Rowe, F. "Tracking spatio-temporal energy vulnerability: A composite indicator for England and Wales". Regional Studies, Regional Science. doi.org/10.1080/21681376.2025.2472069

2025 Tao, S., Rowe, F. and Shan, H. "Nonlinearities and threshold points in the effect of contextual features on the spatial and temporal variability of bus use in Beijing using explainable machine learning: Predictable or uncertain trips?". Journal of Transport Geography. doi.org/10.1016/j.jtrangeo.2025.104126

2024 González-Leonardo, M., Cabrera-Arnau, C., Neville, R., Nasuto, A.. Rowe, F. ""Urban Exodus" During COVID-19 in Mexico? Using digital data to analyze medium-term pandemic impacts on internal population movements." *accepted at Demography* doi.org/10.31219/osf.io/e4au9

2024 Graells-Garrido, E., Rowe, F. "Identifying the spatial patterns of population displacement during wildfires in Valparaíso, Chile." *Regional Studies, Regional Science*. doi.org/10.1080/21681376.2024.2407404

2024 Neville, R., Rowe, F. and Singleton, A. "Unravelling the Brexit-COVID-19 Nexus: Assessing the Decline of EU Student Applications into UK Universities." *Data & Policy*. doi.org/10.1017/dap.2024.42

2024 Nasuto, A. and Rowe, F. "Understanding anti-immigration sentiment spreading on Twitter". *PloS One*. doi.org/10.1371/journal.pone.0307917

2024 Ballantyne, P., Filomena, G., Rowe, F., Singleton, A. "Developing two-dimensional indicators of transport demand and supply to promote sustainable transportation equity." *Computers, Environment and Urban Systems.* 113 doi.org/10.1016/j.compenvurbsys.2024.102179

2024 González-Leonardo, M., Cabrera-Arnau, C., Neville, R., Nasuto, A., Rowe, F. "COVID-19 and internal population movements across the urban hierarchy in Mexico. An analysis using digital data". *Estudios Demográficos y Urbanos* doi.org/10.24201/edu.v39i3.2239

2024 Nelson, T., Frazier, A.E., Kedron, P., Dodge, S., Zhao, B., Goodchild, M., Murray, A., Battersby, S., Bennett, L., Blanford, J.I. and Cabrera-Arnau, C., Claramunt, C., Franklin, R., Holler, J., Lee, A., Mason, S., McKenzie, G., Miller, H., Oshan, T., Rey, S., Rowe, F., Şalap-Ayça, S., Shook, E., Spielman, S., Xu, W. "A research agenda for GIScience in a time of disruptions". *International Journal of Geographical Information Science* doi.org/10.1080/13658816.2024.2405191

2024 González-Leonardo, M., Neville, R., Gil-Clavel, S., Rowe, F. "Where have Ukrainian refugees gone? Identifying potential settlement areas across European regions integrating digital and traditional geographic data." *Population, Space and Place*. doi.org/10.1002/psp.2790

2024 González-Leonardo, M., Rowe, F., Potančoková, M., Goujon, A. "Assessing the differentiated impacts of COVID-19 on the immigration flows to Europe." *International Migration Review*. doi.org/10.31219/osf.io/bsezk

2024 Slocombe, H., Rowe, F. "Representing austerity: Baby Banks and news media." *The Geographical Journal*. 10.1111/geoj.12602

2024 Ward, C., Robinson, C., Singleton, A., Rowe, F. "Spatial-Temporal Dynamics of Gas Consumption in England and Wales: Assessing the Residential Sector Using Sequence Analysis." *Applied Spatial Analysis and Policy*. doi.org/10.1007/s12061-024-09584-9

2024 Houlden, V., Robinson, C., Franklin, R.S., Rowe, F., Pike, A. ""Left Behind" Neighbourhoods in England: Where They Are and Why They Matter." *The Geographical Journal*.

2024 Pelikh, A., Rowe, F. "Increasing diversity, precarity and prolonged periods of education in the transition from school to work in Britain." *Population, Space and Place.*

2024 Shang, L-D, Rowe, F., Lin, E., "Estimating the Causal Impact of Non-traditional Household Structures on Children Educational Performance Using a Machine Learning Propensity Score." *Asia Pacific Education Review*. doi.org/10.1007/s12564-023-09916-3

2024 Neville, R., Rowe, F., Singleton, S. "Nonlinear relationships between human development and international student mobility: The prominent role of employment prospects and cultural and linguistic ties." *Population, Space and Place*, p.e2764.. doi.org/10.1002/psp.2764

2024 Rose, N., Dolega, L., Rowe, F. "Using loyalty card data to understand the impact of weather on click & collect behaviours in UK retailing". *The International Review of Retail, Distribution and Consumer Research*, pp.1-22. doi.org/10.1080/09593969.2024.2316055

2024 Fierro, P., Aravena-Gonzalez, I., Aroca, P., Rowe, F. "Geographies of discontent: measuring and understanding the feeling of abandonment in the Chilean region of Valparaiso (2019–2021)". *Cambridge Journal of Regions, Economy and Society*, p.rsae004. doi.org/10.1093/cjres/rsae004

2023 Calafiore, A., Samardzhiev, K., Rowe, F., Fleischmann, M. and Arribas-Bel, D. "Inequalities in experiencing urban functions. An exploration of human digital (geo-) footprints". *Environment and Planning B: Urban Analytics and City Science*, p.23998083231208507. doi.org/10.1177/23998083231208507

2023 Aparicio-Castro, A., Wiśniowski, A., Rowe, F. "A bayesian approach to estimate annual bilateral migration flows for south america using census data." *Royal Statistical Society. Journal. Series A: Statistics in Society*. doi.org/10.1093/jrsssa/qnad127

2023 Rowe, F., González-Leonardo, M., Champion, T. "Virtual special issue: Internal migration in times of COVID-19." *Population, Space and Place*. doi.org/10.1002/psp.2652

2023 Charles-Edward, E., Bernard, A., Rowe, F., Abel, G. "The Impact of International Migration on Global Population Redistribution, 1990-2015." *International Migration Review*. doi.org/10.1177/01979183231175114

2023 Owen, D., Arribas-Bel, D., Rowe, F. "Tracking the transit divide: a multilevel modelling approach of the relationship between urban inequalities, Covid-19, and public transport ridership disparities." *Sustainability*. doi.org/10.3390/su15118821

2023 Graells-Garrido, E., Opitz, D., Rowe, F., Arriagada, J. "A data fusion approach with mobile phone data for updating travel survey-based mode split estimates." *Transportation research part C: emerging technologies*, 155, p.104285. doi.org/10.1016/j.trc.2023.104285

2023 Tseng, Y.W., Rowe, F. and Lin, E.S. "An Elon Musk generalist or a specialist? The impacts of interdisciplinary learning on post-graduation outcomes". *Studies in Higher Education*, pp.1-26. doi.org/10.1080/03075079.2023.2252889

2023 González-Leonardo, M., Rowe, F., "A 'donut effect'? Assessing housing transactions during COVID-19 across the Spanish urban–rural hierarchy". *Regional Studies, Regional Science*. 10(1), pp.471-472. doi.org/10.1080/21681376.2023.2191684

2023 Rose, N., Rowe, F., Dolega, L. "How consumer behaviours changed in response to COVID-19 lockdown stringency measures: A case study of Walmart" *Applied Geography*. https://doi.org/10.1016/j.apgeog.2023.102948

2023 González-Leonardo, M., Potančoková, M., Yildiz, D., Rowe, F. "Quantifying the impact of COVID-19 on immigration in receiving high-income countries" *Plos One*. https://doi.org/10.1371/journal.pone.0280324

2023 González-Leonardo, M., Newsham, N., Rowe, F. "Understanding Population Decline Trajectories in Spain: A Sequence Analysis Approach." *Geographical Analysis*. https://doi.org/10.1111/gean.12357

2023 Rowe, F., Calafiore, A., Arribas-Bel, D., Samardzhiev, K., Fleischmann, M. "Urban Exodus? Understanding Human Mobility in Britain During the COVID 19 Pandemic Using Meta-Facebook Data." *Population, Space and Place*. https://doi.org/10.1002/psp.2637

2023 Patias, N., Rowe, F., Arribas-Bel, D., "Local urban attributes defining ethnically segregated areas across English cities: A multilevel approach." *Cities*. doi.org/10.1016/j.cities.2022.103967

2023 Thomas, J., Rowe, F. and Lin, E.S. "Declining fertility in Taiwan: the deterring impact of housework imbalance". *Asian Population Studies*, 19(3), pp.270-288. doi.org/10.1080/17441730.2022.2035555

2022 Newsham, N., Rowe, F. "Understanding trajectories of population decline across rural and urban Europe: A sequence analysis." *Population, Space and Place*.

https://doi.org/10.1002/psp.2630

2022 Rowe, F. "Using digital footprint data to monitor human mobility and support rapid humanitarian responses." *Regional Studies, Regional Science*. 9(1), pp.665-668. doi.org/10.1080/21681376.2022.2135458

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2026 Zhang, W., Cleary, E., Rowe, F., Chaudhuri, S., Bondarenko, M., Lai, S., Tatem, A. "Geospatial foundation-model embeddings improve population estimation unevenly across space and scale"10.48550/arXiv.2605.01650

2026 Rowe, F., Soto-Diaz, J., Soto, G. "How Do Local Labor Markets Look from Above? An Automated Satellite-Imagery Approach". Under review. doi.org/10.2139/ssrn.5351760

2026 Rowe, F., Mahony, M., Tao, S., "Assessing Machine Learning for Near-Real Time Bus Ridership Prediction During Extreme Weather." Under review. doi.org/10.48550/arXiv.2204.09792

INVITED TALKS

2025 "Beyond the Immediate Impacts of COVID-19 on Human Mobility". 9 September. Meta. Online. Global.

2025 "Correcting Distorted Human Mobility Data from Mobile Phone Applications". 11 Jul. School of the Environment, University of Queensland, Brisbane, Australia.

2025 "Estimating the isolated and compounded effects of climate change impacts on population displacement". 9 Jul. Mobility Network, Nathan Campus, Griffith University, Brisbane, Australia.

2025 "Correcting for Biases in Location Data from Mobile Phones to Estimate Mobility Flows". 1 Jul. School of Demography, Australian National University, Canberra, Australia.

2025 "Beyond the Census: Leveraging Digital Data to Dynamically Capture Human Mobility and Migration". 26 Jun. Department of Geospatial Sciences and Mathematics, Royal Melbourne Institute of Technology (RMIT), Melbourne, Australia.

2025 "Correcting Distorted Human Mobility Data from Mobile Phones". 24 Jun. School of Geography, University of Melbourne, Sydney, Australia.

2025 "Unlocking the Potential of Mobile Phone and Imagery Data for Understanding Fast-Paced Societal Changes". 24 Jun. Australian Urban Research Infrastructure Network, University of Melbourne, Melbourne, Australia.

2025 "Correcting Distorted Human Mobility Data from Mobile Phones". 20 Jun. School of Civil Engineering, University of Sydney, Sydney, Australia.

2025 "Producing Population-level Estimates of Internal Displacement in Ukraine Using GPS Mobile Phone Data". 23 Jan. Foreign, Commonwealth and Development Office, London, United Kingdom.

2024 "Producing Population-level Estimates of Internal Displacement in Ukraine Using GPS Mobile Phone Data". 22 Oct. CASA seminars, University College London (UCL), London, United Kingdom.

2024 "Using digital trace data for the study of human mobility". 14 May. Urban Beers, University of Chile, Santiago, Chile.

2024 "The persistent impacts of the COVID-19 pandemic on human mobility patterns". 10 May. University Adolfo Ibañez, Viña del Mar, Chile.

2024 "Using Geographic Data Science to study patterns of human mobility". 9 May. University of Chile, Santiago, Chile.

2024 "Unleashing the potential of digital footprint data to understand human mobility: Opportunities, practice and challenges". 2 Apr. University of St Andrews, St Andrews United Kingdom.

2023 "A Data Fusion Approach: Using Mobile Phone Data to Update Travel Survey-based Mode Split Estimates". 10 Nov. Computer Science Department, University of Exeter, Exeter, United Kingdom.

2023 "Oportunidades y desafios del uso de trazas digitales para el análisis de la movilidad y migración humana". Uso de Datos Digitales en el Estudio de la Movilidad y la Migración Humana. 2 Nov. Online, Mexico.

2023 "Digital Footprint Data for Human Mobility: Opportunities, Practice & Challenges". 31 Oct. Jönköping International Business School, Jönköping University, Jönköping, Sweden.

2023 "Identifying Place-based Trajectories of Spatial Inequality Using Sequence Analysis". 10 Mar. Alan Turing Institute-Newcastle University, Newcastle, United Kingdom.

2023 "ITINERANT InequaliTies IN Experiencing uRbAn fuNcTion". 15 Feb. Online, NEtwoRks, Data, and Society (NERDS), IT University of Copenhagen, Copenhagen, Denmark.

2022 "Digital footprint data: Opportunities, Practice and Challenges". 17 Nov. Online, Cedeplar, Universidade Federal de Minas Gerais, Brasilia, Brazil.

2022 "GIS Research Impact." 5 Apr. GISRUK 2020, University of Liverpool, Liverpool, United Kingdom.

2022 "Big Data and Human Geography: Current Practice and Challenges". 25 Mar. Centre d'Estudis Demogràfics (CED), Universitat Autònoma de Barcelona, Barcelona, Spain.

2021 "Data Innovation Event. Harnessing data innovation for migration policy in Europe and Africa State of the art and future directions". 29 Nov. Hybrid, International Organization for Migration (IOM), Berlin, Germany.

2021 "Measuring how 15-minutes urban accessibility shapes human mobility in Barcelona". 15 Nov. Online, Jönköping University, Sweden.

2021 "Geographic Data Science". 2 Nov. Online, Instituto de Data Science, Universidad del Desarrollo, Chile.

2021 "Big Data: Opportunities and Challenges". 14 Oct. Online, Instituto de Data Science, Universidad del Desarrollo, Chile.

2021 "Twitter: Maximising Research Impact Potential" 2021 Royal Statistical Society Conference, Young Statisticians Section of the Royal Statistical Society Special Session. 8 Sep. Manchester, United Kingdom.

2020 "Machine learning in the Social Sciences." Methods Northwest. North West Social Science Doctoral Training Partnership. 12 Nov. Online. United Kingdom.

2020 "Using Machine Learning and Twitter To Profile Attitudes Towards Immigration." Economics Seminar ICESI, 30 July, Universidad Icesi. Cali, Colombia.

2020 "Using Machine Learning and Twitter Data to Profile Attitudes Towards Immigration.", Max Planck Institute for Demographic Research (MPIDR), 10 March, Rostock, Germany.

2019 "Measuring Urban Forest Using Street-Level Imagery and Deep Learning". Building a future for the urban analytics blueprint. The Alan Turing Institute. 17 Oct. Bristol. United Kingdom.

2019 "International Migration: Busting Myths!." Roxby Lecture. University of Liverpool. 16 May. Liverpool. United Kingdom.

2019 "Migration, Cities and Big Data." Population and Development Division, United Nations, 15 April, New York, United States.

2019 "Migration, Urbanisation and Big Data." University of Toronto, 4 April, Toronto, Canada.

2018 "The Impact of Internal Migration on Population Redistribution in Europe." University of Manchester Seminar, 11 May, Manchester, United Kingdom.

2016 "Socioeconomic residential segregation, intrametropolitan spatial inequalities and internal migration in Latin America large cities." Latin American Demographic Centre (CELADE), Economic Commission for Latin America and the Caribbean (CELADE), United Nations (UN), 30 May, Santiago, Chile.

2016 "The returns to migration and human capital accumulation pathways: Non-metropolitan youth in the school-to-work transition." National Institute of Demographic and Economic Analysis, The University of Waikato, 17 May, 2016, Hamilton, New Zealand.

2015 "Careers and migration pathways and outcomes of young people." Research Roundtable, educational, employment and migration pathways and outcomes of young people. Department of Education and Training, 15 October, 2015, Brisbane, Australia.

2015 "Internal migration and development in Latin America." Latin American Demographic Centre (CELADE), Economic Commission for Latin America and the Caribbean (CELADE), United Nations (UN), 14-15 May, Santiago, Chile.

2014 "The effects of economic liberalisation on inter-regional labour migration in a transition economy, Chile." Tinbergen Workshop, 19-20 May, Amsterdam, The Netherlands.

Campus Talks

2024 "Geographic Data Science Lab Partnerships with the United Nations International Organization for Migration." University of Liverpool. 24 Apr. Liverpool. United Kingdom.

2023 "Geographic Data Science Lab. Who we are." University of Liverpool. 12 Jun. Liverpool. United Kingdom.

2023 "Machine Learning & AI projects." University of Liverpool. 8 Mar. Liverpool. United Kingdom.

2022 "Urban Exodus? Human Mobility During COVID-19" University of Liverpool. 8 Sep. Liverpool. United Kingdom.

2020 "Using Twitter to Track Immigration Sentiment During Early Stages of the COVID-19 Pandemic." University of Liverpool. 4 Jun. Liverpool. United Kingdom.

2018 "Regional Science." University of Liverpool. 21 Oct. Liverpool. United Kingdom..

2018 "Totes amaze-balls data opportunity." University of Liverpool. 12 Feb. Liverpool. United Kingdom.

2017 "Why has the distribution of BMI widened? Evidence for England, 2004-2013" University of Liverpool. 22 Mar. Liverpool. United Kingdom.

CONFERENCE ACTIVITY

Plenary Addresses

2025 "From experimental to official migration statistics: A conceptual and actionable framework for quality assurance in mobile-network operator- derived estimates." Conference: Integrating

MNO Data into Official Statistics: challenges and innovation. 23-25 September. OECD, Paris, France.

2025 "Smart Data Research UK panel – "Let's do good things with data"" Royal Statistical Society International Conference 2025. 2 September. Edinburgh, Scotland.

2025 "Beyond the polls: Decoding migration sentiment with AI and digital trace data." Approaches to Migration, Language and Identity. 8 September. Hope University, Liverpool, United Kingdom.

2025 "Releasing the full potential of digital trace data for human migration research: Opportunities, challenges and practice." NERSA/NARSC Program in Mobility, Migration & Regional Science. 5 August. Global. Online.

2025 "Mobile Phone Data for Cross-border Population & Expenditure Flows: Ideas & Challenges." Measurement of the flows across the internal borders of the European Union. Workshop. 14-16 May. Join Research Centre, European Commission, Ispra, Italy.

2025 "Producing Population-level Estimates of Internal Displacement in Ukraine Using GPS Mobile Phone Data." Conference on New Data and Directions in the Study of Migration. 28 February, The University of Hong Kong. Faculty of Social Sciences. Hong Kong.

2025 "Leveraging Digital Data Streams for Internal Population Movements." Symposium Measuring Migration in Latin America: Leveraging digital traces, registers, censuses, and surveys. 25-26 February, Montevideo, Uruguay.

2024 "Panel: Best Practices: Harnessing Mobile Phone Data Related to Migration in the Public Sector." Validation Workshop for the African Union Guidelines and toolkit for the collection and utilization of quality administrative data for labour migration statistics 2024. 2-4 Dec. Petroria, South Africa.

2024 "Interdisciplinary Workshop on Machine Learning and AI Interdisciplinary Workshop on Machine Learning and AI", 2024. 15 Nov. Liverpool, United Kingdom.

2024 "The role of place for sociodemographic analysis and policy making." The critical role of information on "places" and geography to improve social and demographic statistics. The Friends of the Chair (FoC) Group on Social and Demographic Statistics. 2024. 10 Oct. New York, United States.

2024 "Digital Footprint Data for Population Movement." First workshop: Understanding the opportunities, limitations and challenges of digital data sources for nowcasting populations 2024. 26 Sep. Liverpool, United Kingdom.

2024 "Where have Ukrainian refugees gone? Integrating digital and traditional geographic data." International migration in Central and Eastern Europe: changing patterns, determinants, and measurement challenges 2024. 17 Sep. Max Planck Institute for Demographic Research. Rostock, Germany.

2024 "What do we know about the biases and representation in human mobility data extracted from digital platforms?." Symposium on Spatiotemporal Data Science 2024. 23 Jul. Virginia Tech Research Center – Arlington & George Mason Univ. Mason Square, United States.

2024 "Artificial Intelligence for Small Area Level Population Forecast." Population projections for small areas. Methodological contributions and challenges for gender analysis. 4 Jun. El Colegio de México, Ciudad de México, México.

2024 "Unleashing the Potential of Digital Data to Understand Population Movement." XVIII

Congreso de la Población Española y II Congreso Internacional de Geografía de la Población. 12 Jun. Universidad de Granada, Granada, España.

2023 "Enabling Reproducible Spatiotemporal Data Science: Practice & Challenges." The Symposium on Spatiotemporal Data Science. 15 Jul. Harvard University, Cambridge, United States.

2023 "Digital Footprint Data for Human Mobility: Practice & Challenges." British and Irish Regional Science Conference. 4 Jul. Newcastle University, Newcastle, United Kingdom.

2023 "Too Old? Updating Survey Transport Mode Estimates Using Mobile Phone Data." OR Geospatial Data Science Knowledge Exchange Seminar Geospatial Data Science for Future Cities and Society: Focusing on Analysis of Digital Footprint Data. 8 Jun. Korea Research Institute for Human Settlements (KRIHS), Sejong, South Korea.

2022 "Digital Footprint Data for Human Mobility: Practice & Challenges." The First Geospatial Innovation Conference Strengthening African Geospatial. Online, 11 Nov.

2022 "Non-traditional Data to Measure Human Mobility: Progress, Challenges and Practice." Spatial Data Science Conference, 19 May, London, United Kingdom.

2017 "The Rise of International Graduates: Contributing to Regional Australia." International Education Association of Australia 8th Mid-Winter Researchers' Seminar, 4 July, Brisbane, Australia.

2015 "Existing Chilean-Australian relationships: The role of human capital mobility and research/policy implication." Disseminating Knowledge to reinforce Chilean? Australian Networks, 4-5 Sep, Melbourne, Australia.

2015 "The role of human capital mobility in enhancing science." Strengthening ties between Chile and Australia, First Meeting of Chilean Scientist in Brisbane Australian Networks, 31 July, Brisbane, Australia. Keynote presentation.

Sessions Organised

2025 "Workshop: Spatial interaction modelling." Northeastern Regional Science Association (NERSA) Workshop on Mobility, Migration, and Regional Science. NERSA / NARSC Program for Early Career Regional Science Scholars. 6 Jun-13 Aug. Online. Co-organisers: Ting Zhang, Keith Waters. www.regionalscience.org

2023 "Workshop: An Introduction to Network Analysis." GIScience Conference. 12-15 Sep. University of Leeds, Leeds, United Kingdom. Co-organisers: Carmen Cabrera-Arnau, Andrew Renninger, Marin Maureira and Rafael Prieto-Curiel. crmnca.github.io/nets4gis/

2023 "Special Session: Digital footprint data for population science." British Society for Population Studies Conference. 11-13 Sep. Keele University, Keele, United Kingdom.

2021 "Special Session: Human mobility, Migration and Stay-in-Place Patterns and Changes During Pandemic Times." European Regional Science Association Conference. 24-27 Aug. Bolzano, Italy. Co-organisers: Alessandra Faggian and Rachel Franklin.

2018 "Special Session: Doing regional science with new sources of big data." European Regional Science Association Conference. 28-31 Aug. Cork, Ireland.

2018 "Workshop Session: The Potential of Satellite Imagery for Studying Human Activity." Workshop. 14-15 Jun. Liverpool, United Kingdom.

Specialist meetings

2025 "Geospatial Foundation Models: A Collaborative Roadmap" Workshop. 23 October. University of Leeds, Leeds, United Kingdom.

2025 "From experimental to official migration statistics: A conceptual and actionable framework for quality assurance in mobile-network operator- derived estimates." Conference: Integrating MNO Data into Official Statistics: challenges and innovation. 23-25 September. OECD, Paris, France.

2025 "Policy Workshop." (Presenting the Framework and Exploring Broader Contexts) United Nations, International Organization for Migration, Berlin, Germany. May. 21st-22nd 2025.

2025 "Policy Workshop." (Engaging Government Stakeholders for Practical Insights) United Nations, International Organization for Migration, London, United Kingdom. Mar. 25th 2025.

2025 "Policy Workshop." (Shaping a Policy-driven Framework for Displacement Estimates - Webinar) United Nations, International Organization for Migration, London, United Kingdom. Feb. 13th 2025.

2024 "Panel Webinar." (The Sprint of the Friends of Chair (FoC) Group on Social and Demographic Statistics - Webinar) United Nations, Statistics Division Department of Economic and Social Affairs, New York, United States. Oct. 10th 2024.

2024 "Policy Workshop." (Global Data Facility Mobile Phone Data Program for Policy: Cohort 1 Launch Workshop) World Bank, Washington, United States. Oct. 7-10th 2024. global-data-facility-mobile-phone-data-program-for-policy-cohort-1-launch-workshop

2024 "Policy Workshop." (Harnessing Alternative Data for Policy-Driven Assessments of Internal Population Displacement) United Nations, International Organization for Migration, London, United Kingdom. Jul 10th 2024.

2024 "Training Workshop." (Using Digital Footprint Data to Understand Population Movement). El Colegio de México, Ciudad de México, México. June 5th 2024. `fcorowe.github.io/dfd4mobility_mx/`

2024 "Workshop." (Validation of internal migration data [Validación de Datos de Migración Interna]) – Ministry of Social Development and Household, Government of Chile, Santiago, Chile. May 8th 2024.

2023 "Specialist meeting." (Spatial Data Science in an Age of Scientific Disruption) – University of California Santa Barbara (UCSB), Santa Barbara, California, US. December 8-9th 2023. spatial-data-science-age-scientific-disruption

2023 "Training Workshop." (Policy Workshop: RECAST project - Using Digital Footprint Data to Measure and Monitor Human Mobility) – Project Findings. United Nations. Santiago, Chile. July 20th 2023. fcorowe.github.io/dfd4mobility/

2023 "Policy Engagement Workshop." (Policy Workshop: RECAST project - Using Digital Footprint Data to Understand Internal Population Movements in Latin America) – Project Findings. United Nations. Santiago, Chile. July 19th 2023.

2022 "Policy Engagement Workshop." (InequaliTies IN Experiencing uRbAn fuNcTion) – Project Findings. University of Liverpool. Liverpool, United Kingdom. July 21st 2022. `https://youtu.be/lhyytz7ayTY`

2022 "Introduction to Geographic Data Science." BIGSSS Computational Social Science Summer

School on Social Cohesion. University of Groningen. Groningen, The Netherlands. July 5th 2022. http://janlo.de/bigssscss/

2021 "Spatial and Temporal Thinking in Data-Driven Methods." Spatial Data Science Symposium 2021. University of California Santa Barbara. Online. http://sdss2021.spatial-data-science.net

2021 "Towards urban analytics 2.0." The Alan Turing Institute. Leeds, United Kingdom. Nov. 29-30. https://www.turing.ac.uk/events/towards-urban-analytics-20

2021 "Using mobility data in urban science workshop." The Alan Turing Institute. Online, United Kingdom. Oct. 27-28. urban-analytics

2020 "3rd Meeting. The 100 Questions Initiative: Migration cohort" the Big Data for Migration (BD4M) Alliance, International Organization for Migration (IOM), the European Commission and The GovLab. Online. Apr 28 2020. Online. https://migration.the100questions.org

2020 "AI and data science in the age of COVID-19 Workshops". The Alan Turing Institute. Online, United Kingdom. Nov 25-Dec 9. https://www.turing.ac.uk/events/ai-and-data-science-age-covid-19

2019 "Second UK2070 Commission National Symposium". Durham, United Kingdom. Nov 18.

2019 "The 100 Questions Initiative: Migration cohort - 2nd Meeting" the Big Data for Migration (BD4M) Alliance, International Organization for Migration (IOM), the European Commission and The GovLab. Online. Sep. 13 2019. Online. https://migration.the100questions.org

2019 "First UK2070 Commission National Symposium", hosted by Leeds City Council. Leeds, United Kingdom. Jun 13.

2019 "The 100 Questions Initiative: Migration cohort - 1st Meeting" the Big Data for Migration (BD4M) Alliance, International Organization for Migration (IOM), the European Commission and The GovLab. Online. Jun. 27 2019. Online. https://migration.the100questions.org

2014 "Spatial Demography Symposium": Spatial mobility patterns of overseas graduates in Australia. Hosted by the University of Queensland, Brisbane, Australia. Jul 2.

IMPACT & KNOWLEDGE EXCHANGE ACTIVITIES

Recent

2025 "From experimental to official migration statistics: A conceptual and actionable framework for quality assurance in mobile-network operator- derived estimates." Conference: Integrating MNO Data into Official Statistics: challenges and innovation. 23-25 September. OECD, Paris, France.

2025 "Smart Data Research UK panel – "Let's do good things with data"" Royal Statistical Society International Conference 2025. 2 September. Edinburgh, Scotland.

2025 "Policy Workshop." (Presenting the Framework and Exploring Broader Contexts) United Nations, International Organization for Migration, Berlin, Germany. May. 21st-22nd 2025.

2025 "Policy Workshop." (Mobile Phone Data for Cross-border Population & Expenditure Flows: Ideas & Challenges) Joint Research Centre European Commission, Ispra, Italy. May. 14th-16th

2025.

2025 "Policy Workshop." (Engaging Government Stakeholders for Practical Insights) United Nations, International Organization for Migration, London, United Kingdom. Mar. 25th 2025.

2025 "Policy Workshop." (Shaping a Policy-driven Framework for Displacement Estimates - Webinar) United Nations, International Organization for Migration, London, United Kingdom. Feb. 13th 2025.

2024 "Policy Workshop." (Harnessing Alternative Data for Policy-Driven Assessments of Internal Population Displacement) United Nations, International Organization for Migration, London, United Kingdom. Jul 10th 2024.

2024 "Workshop: Artificial Intelligence for Small Area Level Population Forecast." Population projections for small areas. Methodological contributions and challenges for gender analysis. El Colegio de México, Ciudad de México, México. Jun 4th.

2024 "ONS Outreach Engagement Meeting." (The future of population and migration statistics) – Office for National Statistics. Greater Manchester. April 24th 2024.

2024 "Online Meeting." (Learning from continuous surveying and operational use of forced displacement data in war-torn Ukraine) – World Bank Fragility Forum 2024: Adapting and Innovating in a Volatile World Partnership Day. Online. March 1st 2024. `https://events.teams.microsoft.com/event/dae456a2-d168-419a-b605-d0f41953722b@1588262d-23fb-43b4-bd6e-bce49c8e6186`

2024 "Online Meeting." (Modelling accessibility to Early Years Educational providers) – Ofsted, Higher Analytical Officer, Early Years Data and Analysis. Online. February 20th 2024. `https://www.gov.uk/government/organisations/ofsted`

2023 "Online Meeting." (Measuring cumulative impacts) – Equality Trust. Online. October 17th 2023.

2023 "Online Meeting." (Using mobile phone data to estimate population displacement within Ukraine) – Project Findings. United Nations, International Organization for Migration. Online. November 9th 2023. `https://gmdac.iom.int`

2023 "Online Meeting." (Empirical approaches to assessing interventions: Concepts, Data and Methods). Equality Trust. Online. October 17th 2023. `https://equalitytrust.org.uk`

2023 "Policy Engagement Workshop." (Policy Workshop: RECAST project - Using Digital Footprint Data to Understand Internal Population Movements in Latin America) – Project Findings. United Nations. Santiago, Chile. July 19th 2023.

2022 "Policy Engagement Workshop." (InequaliTies IN Experiencing uRbAn fuNcTion) – Project Findings. University of Liverpool. Liverpool, United Kingdom. Jul 22. `https://youtu.be/lhyytz7ayTY`

2021 "An Equal Recovery Inquiry: Geographic Data Science Lab (GDSL) submission". Submission of evidence to the United Kingdom Parliament to inform discussions at the Treasury Committee on An Inquiry on Equal Recovery in the context of COVID-19. Online, United Kingdom. Jun 27. Submission reference: $UXW268417$ `https://committees.parliament.uk/work/1218/an-equal-recovery/`

2021 "Policy Brief: Tracking Sentiment about Immigration at the Early Stages of the COVID-19 Pandemic". Submission to inform policy discussions and recommendations at the

International Organization for Migration, Global Migration Data Analysis Centre and the German Ministry of Interior. Berlin, Germany. Approved: Jul 23.
https://gmdac.iom.int/the-team

2021 "Experts Database: I was included in the United Nations Network Migration Database". Online, United States. Jun 20.
https://migrationnetwork.un.org/hub/experts-database

2021 "Policy Brief: What do 'left behind' areas look like over time? Developing place–based typologies of left behind areas". Submission to inform policy discussions and recommendations at the All-Party Parliamentary Group for 'Left Behind' Neighbourhoods. London, United Kingdom. Approved: Mar 31.
https://www.appg-leftbehindneighbourhoods.org.uk

2021 "2021 Census – how it matters for research". Online, United Kingdom. Mar 15.
https://bit.ly/2WvMknK https://bit.ly/2WvMknK

2021 "2021 Census – why should you fill it in?". Online, United Kingdom. Mar 9.
https://bit.ly/2USknWj https://bit.ly/2USknWj

2021 "Policy Brief: Socio-economic Inequality and Geographic Spread of the COVID-19 Pandemic in England". Submission to inform policy discussions and recommendations at the UK2070 Commission. United Kingdom. http://uk2070.org.uk

2020 "Policy Brief: The Geography of the COVID-19 Pandemic in England. Responding to COVID-19 in the Liverpool City Region". Submission to inform policy discussions and recommendations at the Liverpool City Region. Liverpool, United Kingdom. https://www.liverpool.ac.uk/heseltine-institute/covid-19policybriefs/

2020 "AI and data science in the age of COVID-19 Workshops". Selected as an expert in the use of AI and data science to assess the implications of COVID-19, and contribute evidence to develop a report documenting the lessons during the COVID-19 pandemic and strategic priorities going ahead. Online. United Kingdom.
https://www.turing.ac.uk/events/ai-and-data-science-age-covid-19

2020 "Policy Brief: Changes in Neighbourhood Socio-economic Structures in Teesside, 1971-2011". Submission as part of the Teesside Task Force, part of the work programme of the UK2070 Commission: An Inquiry into a New Economic Vision for the Teesside Economic Region, to inform and develop regional policy recommendations. Online. United Kingdom.
http://uk2070.org.uk/2020/10/28/
uk2070-commission-launches-teesside-taskforce/

2020 "Data Innovation Resource: Estimating Global Migration Flows by Using Machine Learning". Project to showcase how machine learning techniques can be used to produce data and inform policy. Project featured on the Data Innovation Directory of the International Organization for Migration. Berlin, Germany.
https://www.migrationdataportal.org/data-innovation/
global-migration-traditional-innovative-data-machine-learning

2020 "Data Innovation Resource: Using Twitter to Understand Attitudes Towards Immigration". Project to showcase how novel data can be used to inform policy. Project featured on the Data Innovation Directory of the International Organization for Migration. Berlin, Germany.
https://www.migrationdataportal.org/data-innovation/
understanding-public-attitudes-towards-migrants-chile-using-twitter-data

2020 "Data Innovation Resource: Using Twitter to Understand Attitudes Towards Immigration". Project to showcase how novel data can be used to inform policy. Project featured on the Data Innovation Directory of the International Organization for Migration. Berlin, Germany.
https://www.migrationdataportal.org/data-innovation/understanding-public-attitudes-towards-migrants-chile-using-twitter-data

2019 "Bremen International Graduate School of Social Sciences (BIGSSS) Summer School in Computational Social Science on Social Cohesion in Groningen, Netherlands". Selected as an expert and project leader to mentor and develop a group of young academics to develop and complete a research project during two weeks.
http://bigsss-css.jacobs-university.de

2019 "The 100 Questions – Migration Domain". The 100 Questions Initiative sought to map the world's 100 most pressing, high-impact questions that could be answered if relevant datasets were leveraged in a responsible manner. Selected as a bilingual expert to contribute to formulate and vote for the most relevant questions related to migration research. Project led by the GovLab in collaboration with the European Commission, International Organization for Migration and Global Migration Data Analysis Centre.
https://migration.the100questions.org

2019 "R package CIM: Compositional Impact of Migration reached over 2,300 downloads". The package produces statistical indicators of the impact of migration on the socio-demographic composition of an area.
https://cran.r-project.org/web/packages/CIM/index.html

2019 "Findings from my research on internal migration decline across Europe featured in an Institute for Public Policy Research report". the Divided and Connected – Regional inequalities in the North, The UK and the developed world state of the north 2019.
https://www.ippr.org/files/2019-11/sotn-2019.pdf

2019 "Findings from my research on the impacts of internal migration on the population structures of Latin America referenced in a UN expert group meeting 'Commission on Population and Development' at the fifty-first session on 'Sustainable Cities, Human Mobility and International Migration'".
https://www.un.org/development/desa/pd/events/CPD51

TEACHING

Postgraduate

2023– Population Science. https://www.population-science.net

2020– Spatial Modelling for Data Scientists. https://gdsl-ul.github.io/san

2016– Geographic Data Science Dissertation.

2016–2017 Population, Health and Mortality.

2016–2018 Theorising Human Geography.

Undergraduate

University of Liverpool, United Kingdom

2016– Geography undergraduate dissertation supervisions.

2016– Geography Year 1 and 2 Tutorials.

2018– Toronto Field Class Year 3.

2019–2020 Exploring the Social World.

2016–2021 Population and Societies.

2021–2022 Virtual Field Class Year 3.

2017–2018 Singapore Field Class Year 3.

2016–2019 Statistics for Social Scientists.

2016–2017 Human Geography through Merseyside.

2016–2017 Glasgow Field Class Year 2.

University of Queensland, Australia

2016 Global Population Issues.

2015 Human Mobility and Migration.

2014 Human Settlements.

Universidad Catolica del Norte, Chile

2008 Statistics.

2008 Macro-economics.

Universidad Santo Tomas, Chile

2007 Algebra

2007 Macroeconomics

SUPERVISION

Postdoctoral Research

2025– Susie Deng. "Social Health Hub". The University of Liverpool.

2025– Martina Pardy. "Imago Data Service for Imagery". The University of Liverpool.

2023–24 Michael Mahony. "TIMELY - Validating an up-to-date dataset of internal migration to inform local and regional-scale spatial planning strategies in Chile". The University of Liverpool.

2021–22 Alessia Calafiore. "InequaliTies IN Experiencing uRbAn fuNcTion (ITINERANT)". The University of Liverpool.

Research Assistance

2026– Rodgers Iradukunda. "EXHALE". The University of Liverpool.

2026– Emiliano Beltran. "EXHALE". The University of Liverpool.

2024– Andrea Nasuto. "Dynamic Estimates of Displacement in Conflict Regions: A Policy-driven framework triangulating data". The University of Liverpool.

2025– Matt Mason "Dynamic Estimates of Displacement in Conflict Regions: A Policy-driven framework triangulating data". The University of Liverpool.

2024–24 Matt Mason "Assessing the long- and short-terms impacts of climatic changes on internal population displacement". The University of Liverpool.

2024–24 Andrea Nasuto. "Harnessing Satellite and GPS Phone Data for Policy-Driven Assessment of Internal Population Displacement in Ukraine". The University of Liverpool.

2024–24 Rodgers Iradukunda. "Harnessing Satellite and GPS Phone Data for Policy-Driven Assessment of Internal Population Displacement in Ukraine". The University of Liverpool.

2024–24 Rodgers Iradukunda. "Using mobile phone data to estimate internal population displacement in Ukraine during the start of the Russian invasion". The University of Liverpool.

2023–23 Andrea Nasuto. "RECAST – Strengthening policy collaboration with the United Nations: Recovery and resilience through digital footprint data". The University of Liverpool.

2023–23 Ruth Neville. "RECAST – Strengthening policy collaboration with the United Nations: Recovery and resilience through digital footprint data". The University of Liverpool.

2022–22 Ruth Neville. "Understanding the mobility patterns of Ukrainians Using Twitter". The University of Liverpool.

2021–21 Niall Newsham. "What do "left behind" areas look like over time?". The University of Liverpool.

2020–20 Michael Mahony. "Analysis of Twitter data to measure anti-migration sentiment during COVID- 19". The University of Liverpool.

2019–21 Michael Mahony. "Assessing Machine Learning for Near-Real Time Bus Ridership Prediction During Extreme Weather". The University of Liverpool.

Current PhD supervision

I am currently supervising 6 PhD students.

2025– Mark Van Eker. "Assessing and determining the present and future offer of early childhood education". The University of Liverpool.

2024– Emiliano Beltran Alipio. "Safeguarding urban communities: Using data science to determine the impact of the spatial structure of urban activities and residential segregation on the spreading of infectious diseases". The University of Liverpool.

2023– Rodgers Iradukunda. "Long-term spatial inequalities". The University of Liverpool.

2022– Matt Mason. "Fuelling Hate: The Role of Social Media in Hate Crime". The University of Liverpool.

2022– Matt Howard. "Spatial Inequalities and patterns of Over Indebtedness". The University of Liverpool.

Complete PhD supervision

I have supervised 17 PhD students to completion.

2022–2026 Andrea Nasuto. "Understanding the role of social media in promoting anti-migration sentiment and hate crime". The University of Liverpool. Viva completed in August 2026; PhD awarded in September 2026.

2018–2025 Danial Owen. "Urban sensors: exploiting the opportunities". The University of Liverpool.

2020–2025 Ruth Neville. "Exploring flows of international students into the UK using UCAS Application Data". The University of Liverpool.

2020–2025 Cameron Ward. "Examining the patterns and trends of energy poverty". The University of Liverpool.

2019–2024 Michael Mahony. "Understanding and Predicting the Long-term Labour Market and Migration Trajectories of Immigrants and Their Children in the United Kingdom". The University of Liverpool.

2022–2024 Hannah Slocombe. "Beyond Food Banks: Everyday Geographies of Coping in Austere Times". The University of Liverpool.

2019–2024 Niall Newshaw. "Understanding the extent, causes, consequences and future of population decline in Europe". The University of Liverpool.

2018–2023 Natalie Rose. "Quantifying the impact of weather on high street retail purchases.". The University of Liverpool.

2018–2023 Andrea Apacio. "Modelling and forecasting the spatial and temporal patterns of bilateral international migration flows". The University of Manchester.

2018–2023 Ya-Weng Tseng, "Evaluating Interdisciplinary Learning". National Tsing-Hua University - The University of Liverpool.

2017–2021 Nikos Patias. "Scalable Analytical Framework for Spatio-temporal Data Analysis". The University of Liverpool.

2017–2021 Jac Thomas. "Facilitating Childbearing in Taiwan: The Role of Domestic Gender Equity and Parental Leave". National Tsing-Hua University - The University of Liverpool.

2018–2021 Li-Dan Shang. "The Education System and Labour Market Integration of Immigrants". National Tsing-Hua University - The University of Liverpool.

2014–2020 Angelina Zhi Rou Tang, "Overseas graduates in Australia: Characteristics, dynamics and change". The University of Queensland.

2016–2019 Leo Singer. "Life-course patterns of multimorbidity in its social contexts". The University of Liverpool.

2016–2018 Alina Pelikh. "Transition to adulthood in England and Wales: The analysis of life trajectories of young adults", The University of Liverpool.

2013–2015 Wilma Terblance, "Population Estimates and Projections for Australia's Very Elderly Population at State and National Level", The University of Queensland.

MSc supervision

I have supervised 9 MSc dissertations.

2023–24 Suganthan Parthiban. "To assess the extent to which stringency restrictions influenced human mobility patterns and thus contribute to contain the spread of COVID-19 infection and how socio-economic groups these patterns vary across.". The University of Liverpool.

2022–23 Rodgers Iradukunda. "Using Smartphone Geolocation Data to Understand Displacement and Mobility Patterns of the Ukraine-Russian Conflict". The University of Liverpool.

2018–22 Melanie Green. "Automating detailed urban data extraction from high resolution aerial imagery". The University of Liverpool.

2020 Andrea Nasuto. "Determining the key sources, speed and evolution of the immigration debate

on Twitter in the UK", The University of Liverpool.

2020 Megan Clark. "A quantitative analysis of anti-migrant sentiment on Twitter during the COVID-19 pandemic", The University of Bristol.

2020 Sultan Alotaibi. "COVID-19 Confirmed Cases and Deaths in United Kingdom and Association with Socio-Demographic Profile: A Spatial and Temporal Analysis", The University of Liverpool.

2019 Michael Mahony. "Using satellite imagery and machine learning to estimate access to improved water sources: a case study of Malawi", The University of Liverpool.

2018 Emma Branagan. "The Effects of Social Media on Trading Outcomes: A Company Case Study", The University of Liverpool.

2017 Niall Newsham. "A Projection of the Demographic Impact of Syrian Migration in Germany". The University of Liverpool. *Dissertation awarded the Bob Woods Postgraduate Dissertation Prize in population studies and reworked into a collaborative paper to be published in the Journal of Geographical Systems.*

Honours

2013 Angelina Zhi Rou Tang. "Spatial mobility patterns of overseas graduates in Australia", The University of Queensland.

2011 Rachel Ong. "Human capital and migration of international graduates in Australia: Investigating the impact of the graduate visa", The University of Queensland.

PhD external examination

I have examined 8 PhD theses.

2026 Édith Darin, "Estimating subnational population in data-scarce contexts", University of Oxford.

2026 Sebastian Schmidt, "From Web Text to Spatial Insights: Integrating Organisational Websites and Social Media with Complementary Spatial Data for Socio-Ecological Analyses", Paris Lodron University of Salzburg.

2026 Ademola Adewole, "Using georeferenced building-level administrative data to enhance settlement classification methods for population mapping", University of Southampton.

2024 Alice Battiston, "Enhancing urban liveability and sustainability through geospatial large-scale data analytics", University of Turin.

2024 Peter Dorey, "Understanding Spatial Variation in Fertility An application of Spatial Statistical Methods to Fertility in Great Britain", University of St Andrews.

2023 Sofia Gil-Clavel, "Studying Social Interactions Using Digital Trace Data", University of Groningen.

2023 James Watson "Social and spatial inequalities in healthcare use and health outcomes among people living with dementia in England", University of Liverpool.

2022 Miguel González-Leonardo, "Internal migration by educational attainment in Spain", Centre d'Estudis Demogràfics (CED), Universitat Autònoma de Barcelona.

SERVICE

Journal reviews

Applied Economics Letters • Advances in Life Course Research • Applied Geography • Applied Spatial Analysis and Policy • Australian Population Studies • Big Data & Society • Cities • Comparative Population Studies • Computers, Environment and Urban Systems • Demography • Demographic Research • Economic Geography • EPJ Data Science • Environment and Planning B • Environment and Planning C • Forests • Geographical Analysis • Habitat International • Humanities and Social Sciences Communications • Geojournal • Journal of Geographical Systems • Humanities & Social Sciences Communications • The Australian Economic Review • The Australian Journal of Agricultural and Resource Economics • The Australian Economic Review • The Journal of Annals of Regional Science • Journal of Geographical Systems • Journal of Ethnic and Migration Studies • Journal of Regional Science • Journal of Population Research • Journal of the Royal Society Interface • Journal of Urban Technology • Papers in Regional Studies • PloS One • PNAS • PNAS Nexus • Population and Development Review • Population Research and Policy Review • Population, Space and Place • Progress in Human Geography • Regional Studies • Royal Society Open Science • Regional Science, Regional Studies • NPJ Urban Sustainability • Nature Cities • Nature Communications • Nature Human Behaviour • Scientific Data • Scientific Reports • Socio-Economic Planning Sciences • Spatial Economic Analysis • Sustainable Cities and Society • The International Journal of Human Resource Management • Transportation Research Part C • Transportation Research Part D • Quality & Quantity • UCL Open Environment

Grant review

Austrian Science Fund • Australian Research Council (ARC) • Caixa – Social Research Call • Chilean Intelligence National Agency • Chilean National Commission for Scientific and Technological Research (CONICYT) • European Research Council • Economic Social Research Council (ESRC) • National Science Centre Poland • Wellcome Trust.

Book review

Routledge • Sage.

Professional Membership

2022– Fellow, Royal Geographical Society.

2017– Member, British Society for Population Studies.

2016– Member, Regional Studies Association.

2014– Member, Regional Science Association International.

2013– Member, Chilean Society of Regional Science.

Scientific Software

2025 Mahony, M., Rowe, F., Cabrera-Arnau, C., hhmR - Hierarchical Heatmaps, R package: `https://cran.r-project.org/web/packages/hhmR/index.html`

2021 Patias, N. and Rowe, F., extRatum, R package: `https://CRAN.R-project.org/package=extRatum`

2019 Rowe, F., Patias, N. and Rodríguez-Vignoli, J., Compositional Impact of Migration (CIM), R package: `https://CRAN.R-project.org/package=CIM`

Open data products

2021 Left behind areas in England: https://bit.ly/3aHs4TD

2021 Neighbourhood trajectories in Great Britain 1971-2011: https://data.cdrc.ac.uk/dataset/neighbourhood-trajectories-great-britain-1971-2011

2017 The Chilean Internal Migration (CHIM) database: https://s3.amazonaws.com/geoda/data/CHIM

2017 Functional Labour Market Areas: https://geodacenter.github.io/data-and-lab//FLMA/

Updated October 2026